

The Topology of $\mathbb{R}$

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Open Sets

Definition

A set $\mathcal{U} \subseteq \mathbb{R}$ is **open** if $\forall x \in \mathcal{U}, \exists \delta > 0$ such that $V_\delta(x) := (x - \delta, x + \delta) \subseteq \mathcal{U}$

- Examples are : $\mathbb{R}, (a, b), \phi$
- Non-examples are: $[a, b], [a, b), [a, \infty)$

Theorem

- i) If $\{ U_i \}$ is any collection of open sets , then $\bigcup_i U_i$ is also an **open set** .
- ii) If $\{ U_1, \dots, U_n \}$ is a finite collection of open sets , then $\bigcap_{i=1}^n U_i$ is also an **open set** .

Closed Sets

Definition

A set $\mathcal{U} \subseteq \mathbb{R}$ is **closed** if $\mathcal{U}^c := \mathbb{R} - \mathcal{U}$ is *open*.

- Examples are : $\mathbb{R}$, ϕ , $[a, b]$
- Non-Examples are: $[a, b)$, (a, b)

Definition

A point x is a **limit point of a set** $\mathcal{U}$ if $\exists (a_n) \subseteq \mathcal{U} - \{x\}$ such that $\lim(a_n) = x$

- Example: 0 is a limit point of $(0, 1)$ take $a_n = \frac{1}{1+n}$. Notice that $a_n \in (0, 1) - \{0\}$, and $\lim a_n = 0$.

Definition

A point $x \in \mathcal{U}$ is an **isolated point of** $\mathcal{U}$ if it is not a limit point of $\mathcal{U}$

Closed Sets and Limit Points

Theorem

A set $\mathcal{U}$ is closed iff contains all of its limit points

- proof : ($\Rightarrow$) assume for contradiction that $\mathcal{U}$ is closed and $\exists(a_n) \subseteq \mathcal{U}$ s.t. $\lim a_n = x$ for $x \notin \mathcal{U}$, so $x \in \mathcal{U}^c$. Since $\mathcal{U}^c$ is an open set, $V_\delta(x) \subseteq \mathcal{U}^c$ for some $\delta > 0$. We have also that $|a_N - x| < \delta$ for suitable $N \in \mathbb{N}$ which implies $a_N \in \mathcal{U}$ which is a contradiction.
- ($\Leftarrow$) By contrapositive, let $\mathcal{U}$ is Not closed set, so $\mathcal{U}^c$ is not open which means that there is $x \in \mathcal{U}^c$ such that $\exists a \in V_\delta(x) - \mathcal{U}^c$ for every $\delta > 0$. In particular consider $a_n \in \mathcal{U}$ for $\delta := \frac{1}{n}$. Hence $\lim a_n = x$ for some $x \notin \mathcal{U}$ which means x is a limit point but not in $\mathcal{U}$ $\square$

Open Covers

Definition

Let $\mathcal{U}$ be a set .

- The sets $\{ \mathcal{O}_i \}_{i \in I}$ are **a cover of $\mathcal{U}$** if $\mathcal{U} \subseteq \bigcup_{i \in I} \mathcal{O}_i$
- If each of $\mathcal{O}_i$ is open , then $\{ \mathcal{O}_i \}_{i \in I}$ is **an open cover of $\mathcal{U}$** .
- If $\{ \mathcal{O}_i \}_{i \in I}$ has a *finite* subset $\{ \mathcal{O}_i \}_{i \in N}$ which also a cover of $\mathcal{U}$, then $\{ \mathcal{O}_i \}_{i \in N}$ is called a **finite subcover of $\mathcal{U}$**
- Examples: $(0, 4) \subseteq \{ (\frac{1}{n}, 4 - \frac{1}{n}) \}_{n=1}^{\infty}$ is an open cover .
- $[0, 4] \subseteq \{ (\frac{1}{n}, 4 - \frac{1}{n}) \}_{n=1}^{\infty} \cup (-0.2, 0.2) \cup (3.9, 4.1)$ is an open cover .

Compact Sets

Definition

A set $\mathcal{U}$ is called **compact** if every open cover of $\mathcal{U}$ contains a finite subcover of $\mathcal{U}$.

- Example : $(0, 4)$ is **not** compact set , since the open cover $\{ (\frac{1}{n}, 4 - \frac{1}{n}) \}_{n=1}^{\infty}$ has no finite subcover.

The Heine-Borel theorem

A set $\mathcal{U} \subseteq \mathbb{R}$ is **compact** iff $\mathcal{U}$ is closed and bounded.

- Example: $[a, b]$ is a compact set and $\mathbb{R}$ is **not** compact .

Corollary

A set $\mathcal{U} \subseteq \mathbb{R}$ is **compact** iff every sequence in $\mathcal{U}$ has a subsequence that converges to a limit in $\mathcal{U}$.

Compact Sets

Theorem

Let $\mathcal{U} \subseteq \mathbb{R}$. Then the following are equivalent.

- i) $\mathcal{U}$ is compact.*
- ii) $\mathcal{U}$ is closed and bounded.*
- iii) if (a_n) is a sequence of numbers in $\mathcal{U}$, then there is a subsequence a_{n_k} that converges to a point in $\mathcal{U}$.*

Continuous Functions on Compact sets

Theorem

let $f : A \rightarrow \mathbb{R}$ be continuous function on A . If $\mathcal{U} \subseteq A$ is **compact** , then $f(\mathcal{U})$ is **compact** also.

- proof: let (y_n) be an arbitrary sequence in $f(\mathcal{U})$, then for each $n \in \mathbb{N}$ we have $x_n \in \mathcal{U}$ with $f(x_n) = y_n$. This yields that we have a sequence $(x_n) \subseteq \mathcal{U}$. Since $\mathcal{U}$ is compact , there exists a convergent sequence $(x_{n_k}) \rightarrow x$ for some $x \in \mathcal{U}$. since f is continuous ,
 $f(x_{n_k}) = (y_{n_k}) \rightarrow f(x) \in f(\mathcal{U})$. $\square$
- Example : $f(x) := \sin(x)$, $f([0, \pi]) = [0, 1]$
- Non-example : $g(x) : e^x$, $g(0, 1) = (1, e)$